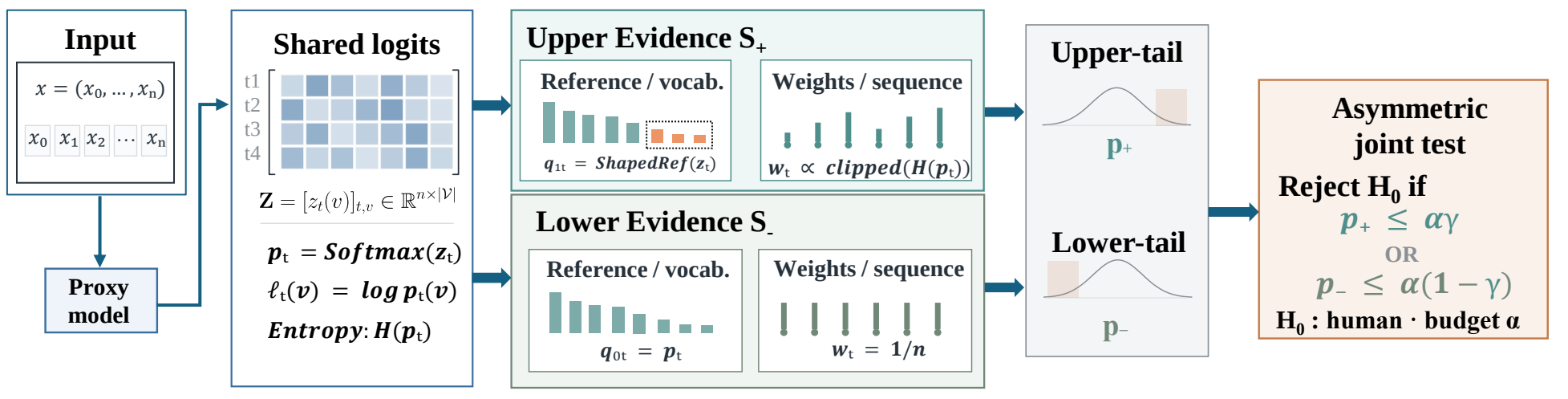

Input
$x = (x_0, \ldots, x_n)$
x_0 x_1 x_2 $\cdots$ x_n
Proxy model
Shared logits
t1
t2
t3
t4
$\mathbf{Z} = [z_t(v)]_{t,v} \in \mathbb{R}^{n \times |\mathcal{V}|}$
$p_t = Softmax(z_t)$
$\ell_t(v) = log\, p_t(v)$
Entropy: $H(p_t)$
Upper Evidence S_+
Reference / vocab.
$q_{1t} = ShapedRef(z_t)$
Weights / sequence
$w_t \propto clipped(H(p_t))$
Lower Evidence S_-
Reference / vocab.
$q_{0t} = p_t$
Weights / sequence
$w_t = 1/n$
Upper-tail
p_+
Lower-tail
p_-
Asymmetric joint test
Reject H_0 if
$p_+ \leq \alpha\gamma$
OR
$p_- \leq \alpha(1-\gamma)$
H_0 : human · budget α